Höhere Technische Lehranstalt Shkoder
Abteilung Systemtechnik – Schwerpunkt Data Science

Schuljahr 2025/26

VHGRAD

Entwicklung eines didaktischen Deep-Learning-Frameworks
mit eigener Autograd-Engine

Diplomarbeit

Verfasst von

Virvi Huta

Betreuer

Unknown

Shkoder, März 2026

Technical Progression of the vhgrad Framework

1. **Foundations of Neural Networks** Introduction to neural network concepts, historical context, and core components such as neurons, layers, and parameters.
2. **Initial Implementations** Manual implementation of single neurons and multi-neuron layers with emphasis on mathematical foundations.
3. **Vectorized Computation** Transition to NumPy-based implementations using vectors, matrices, batching, and efficient matrix operations.
4. **Multi-Layer Network Construction** Encapsulation of dense layers into reusable classes and structured handling of training data.
5. **Activation Functions** Implementation of linear and non-linear activation functions, including Sigmoid, ReLU, and Softmax.
6. **Loss Functions and Evaluation** Integration of categorical cross-entropy loss and accuracy metrics for model evaluation.
7. **Optimization and Backpropagation** Gradient-based optimization, derivatives, chain rule, and full backpropagation pipeline.
8. **Optimizers** Implementation of SGD, learning rate decay, momentum, AdaGrad, RMSProp, and Adam.
9. **Regularization Techniques** L1 and L2 regularization, dropout, and validation strategies to improve generalization.
10. **Classification and Regression** Support for binary classification and regression tasks with appropriate loss functions.
11. **Model Abstraction** Introduction of a unified model object encapsulating layers, losses, and optimizers.
12. **Real-World Data Integration** Data loading, preprocessing, shuffling, batching, and training on real datasets.
13. **Model Persistence and Inference** Saving and loading model parameters and enabling prediction on unseen data.